Examples: $\{0\} \xrightarrow{i} \mathbb{A}^1 \xrightarrow{j} \mathbb{G}_m$

① $D_{\mathbb{G}_m} \subset DMod(\mathbb{G}_m)$ is coherent

$j_* D_{\mathbb{G}_m}$ is not.

$\partial \frac{1}{x} \longrightarrow \frac{-1}{x^2} + \underbrace{\frac{1}{x}\partial}_{=0}$

$D(\mathbb{G}_m)$ is not finitely gen over $D(\mathbb{A}^1)$

$\ni 1, x^{-1}, x^{-2}, x^{-3}, \ldots$

$j_* \mathcal{O}_{\mathbb{G}_m}$ is ID finitely generated over $D(\mathbb{A}^1)$

$\frac{1}{x} \xrightarrow[\frac{1}{dx}]{} \frac{-1}{x^2} \xrightarrow[\frac{d}{dx}]{} \frac{2}{x^3} \longrightarrow \cdots$

② $i^! D_{\mathbb{A}^1}$ is not finitely generated over $D(\{0\}) = k$

$\cong$

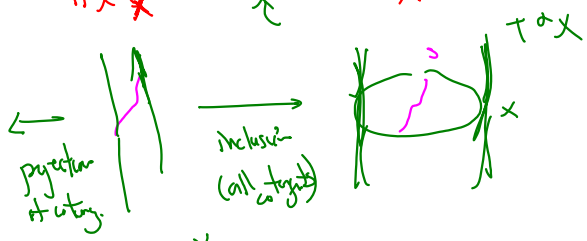
$k[\partial]$ infinite dim.

Lemma: Let $i: S \hookrightarrow X$ be locally closed immersion (i.e. if i is affine, then i_* is exact. it's closed imm. followed by an open imm.)

Then, i_* preserves holonomic $\underset{\text{complex}}{D}$-mod. ($i$ open, i closed)

Proof ① i is closed. $\Rightarrow$ singular support.

$T^*_S \longleftarrow T^*_X X \times_X S \longrightarrow T^*_X X$



w/ kernel $= \mathbb{N}^\vee_{S/X}$

$\cup\!\!\mid \longrightarrow \dim(S) + \dim(\mathbb{N}^\vee) \longrightarrow \dim(X)$

$SS(M)$ $\quad = \dim(X)$ $\quad SS(i_* M)$

$\dim = \dim(S)$

i always affine, i_* is exact (long time ago... Kashiwara)

$i_*: DMod(S) \xrightarrow{\sim} DMod_S(X)$

$\Rightarrow$ exact

② i is open

• assume M is just a module (i.e. in degree 0)

$D(DMod(X)) \xrightarrow{\sim} D_h(DMod(X))$

use truncation

- If i is open, $i_+\mathcal{M}$ is the same as $\mathcal{O}$-push

$$i_+\mathcal{M} = Ri_+\left(\mathcal{M} \overset{L}{\underset{D_S}{\otimes}} \mathcal{O}_S \underset{i^{-1}\mathcal{O}_X}{\otimes} i^{-1}D_X\right)$$

$$= Ri_+\left(\mathcal{M} \overset{L}{\underset{D_S}{\otimes}} \mathcal{O}_S \underset{\mathcal{O}_S}{\otimes} D_S\right) = Ri_+\left(\mathcal{M} \overset{L}{\underset{D_S}{\otimes}} D_S\right)$$

$$= Ri_+(\mathcal{M})$$

diff op on X act thru restriction

<u>Reduce</u>: i is affine by taking open affine cover of S in X, use Čech complex

i.e. assume that i is affine. ($\overset{\mathcal{O}\text{-push}}{\Longrightarrow}$ exact)

<u>Problem</u>: $\mathcal{M}$ is holonomic, i is affine open, then $i_+\mathcal{M}$ is holonomic.

<u>Ex</u>: $i_+\mathcal{O}_{\mathbb{A}^1}$ Why is this what?

$$\frac{1}{x} \xrightarrow{\partial} \frac{1}{x^2} \xrightarrow{\partial} \cdots$$

Does this happen in general?

<u>More reductions</u>: ① $\mathcal{M}$ is generated by m_0 over $D_{\mathbb{A}^1}$

② assume that $\mathcal{O}(U) = \mathcal{O}(X)[\frac{1}{f}]$

<u>Problem</u>: Show $\frac{1}{f^n} m_0 \in D(X) \cdot m_0$ ← not $D(U)$

<u>Lemma</u>: (b-functions) t formal variable

$D(t) \in D_X[t]$ $\quad b(t) \neq 0 \in k[t]$ s.t.

① Why does this help?

② Why is this necessary?

$$b(n) f^n \cdot m_0 = D(n) f^{n+1} m_0$$

$\frac{1}{f} m_0 \in D(X) m_0$

① $n = -1$. $\quad b(-1) \frac{1}{f} m_0 = D(-1) \cdot m_0$

$\frac{1}{f^2} m_0 \in D(X) m_0$

$b(-2) \frac{1}{f^2}(m_0) = D(-1) \cdot \left(\frac{1}{f} m_0\right)$

<u>BUT</u>: assuming $b(-1), b(-2), b(-3) \neq 0$!

Replace m_0 with $\frac{1}{f^r} m_0$ s.t. all the zeros of b are avoided. ②

Pf: $K = \mathrm{Frac}(k[t]) = k(t)$

Base change everything to K.

$\mathcal{U}_K \hookrightarrow X_K$

as $\mathcal{O}$-mod $\simeq K \otimes_k M$

$M_K := f^t K \otimes_k M$

$\theta \cdot (f^t \otimes m)$

$= \frac{t f^t}{f}(\theta \cdot f) m + f^t \theta \cdot m$

Explain: Check that this is action.

Use: $[\theta, f^t] = \frac{t f^t}{f} \cdot \theta(f)$

Extension lemma:

if i open, M holonomic,

$H \subseteq i_* M$ holonomic submodule s.t. $H|_{\mathcal{U}} = M$.

Claim $H \subseteq M_K$ ← pushed to X

M_K / H supp on $X - \mathcal{U}$ i.e. f acts by torsion

i.e. there is r s.t. $f^r \cdot m_0 \in H$

Replace m_0 with $f^r m_0$ i.e. assume $m_0 \in H$

Because H is holonomic: finite length

$D(X_K) \cdot f^t m_0 \supseteq D(X_K) \cdot f^{t+1} m_0 \supseteq \cdots \supseteq \boxed{D(X_K) \cdot f^{t+r} m_0 = D(X_K) \cdot f^{t+r+1} m_0}$

w/o Frac get stuck here

i.e. there is D_0 s.t.

$D_0 \cdot f^{t+1+r} m_0 = f^{t+r} m_0$

Replace $f^{t+r} m_0$ with $f^t m_0$

$D_0 \cdot f^{t+1} m_0 = f^t \cdot m_0$

$D_0 \in D(X) \otimes_k K$

$D_0 = \frac{D(t)}{b(t)}$ ← polynomial

$$\boxed{D(t) \cdot f^{t+1} m_0 = b(t) \cdot f^t m_0}$$

sub $t = u$.

$\boxed{H = M_K \text{ is holonomic}}$

Lemma (extension): $M \in D\mathrm{Mod}(X)$, $j: \mathcal{U} \hookrightarrow X$ open

If $M|_{\mathcal{U}}$ is holonomic, then there is $\mathcal{H} \subseteq M$ s.t.

$\mathcal{H}|_{\mathcal{U}} = M|_{\mathcal{U}}$, and $\mathcal{H}$ is holonomic.

Proof · First coherent extension (exercise: same as in AG)

· i.e. assume $\mathcal{M}$ is coherent

· $\mathbb{D}_X(\mathcal{M})$ lives in degrees $[-\dim(X), 0]$. i.e. $H^0(\mathbb{D}_X\mathcal{M})$ is <u>quotient</u>.

· Roos thm: $\dim SS(H^0(\mathbb{D}_X\mathcal{M})) \le \dim(X)$ i.e. holonomic.

· $\mathbb{D}_X(H^0\mathbb{D}_X\mathcal{M})$ is holonomic in degree 0, <u>submodule</u> of $\mathcal{M}$

· Verify: $\underbrace{\mathbb{D}_X(H^0\mathbb{D}_X\mathcal{M})}_{\text{submodule.}}|_U = \mathcal{M}|_U$, holonomic, ∎

Finish proof: i.e. prove holonomicity.

$\mathcal{M}|_U$ is holonomic: $D_0, \dots, D_r$ generate $\mathcal{A}_m(f^t m_0)$

idea: specialize t. have denominators. Choose $n \in \mathbb{Z}$ avoiding denominators

$\Rightarrow \underbrace{D_0(n), \dots, D_r(n)}_{\text{cut out singular supp.}}$ generate $\mathcal{A}_m(f^n m_0)$ on $D(X)$ ← generator.

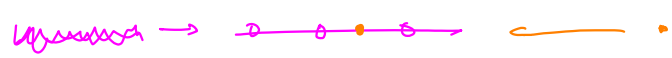
Specialization commutes w/ SS. ... dim preserved, so $\mathcal{M}$ holonomic. ∎

$U_K \hookrightarrow X_K$ ⇝ $\mathcal{M}_K$ lift

Formally: $U_K \hookrightarrow X_K$ → $\mathrm{Spa}\,K$

$\mathrm{Spa}(K) \hookrightarrow \{\mathrm{Spec}\,k[[t]]\} - \{\text{bad points}\} \leftrightarrow \{\text{pt}\}$

$\{\eta\} \longleftarrow \cdot$



generic pt. $\mathbb{A}^1$